EXP-16 : To generate and rank candidate substitution-cipher plaintexts.

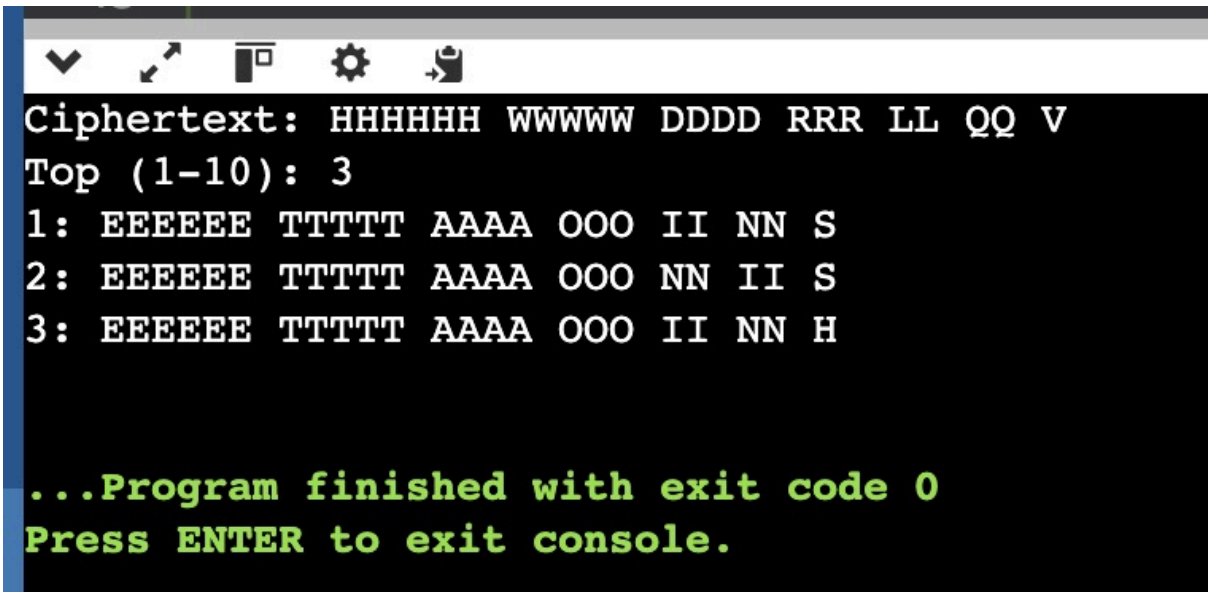
PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>
#include <math.h>
int main(void) {
 char c[501], rank[]="ETAOINSHRDLCUMWFGYPBVKJXQZ";
 double f[]={8.167,1.492,2.782,4.253,12.702,2.228,
 2.015,6.094,6.966,.153,.772,4.025,2.406,6.749,
 7.507,1.929,.095,5.987,6.327,9.056,2.758,.978,
 2.360,.150,1.974,.074}, score[326];
 int count[26]={0}, order[26], top, n=1;
 char key[326][26];
 printf("Ciphertext: ");
 if (!fgets(c,sizeof c,stdin)) return 1;
 printf("Top (1-10): ");
 if (scanf("%d",&top)!=1 || top<1 || top>10) return 1;
 for (int i=0;c[i];i++) {
 if (c[i]>='A' && c[i]<='Z') count[c[i]-'A']++;
 }
 for (int i=0;i<26;i++) order[i]=i;
 for (int i=0;i<26;i++) for (int j=i+1;j<26;j++)
 if (count[order[j]]>count[order[i]]) {
 int t=order[i]; order[i]=order[j]; order[j]=t; }
 for (int i=0;i<26;i++) key[0][order[i]]=rank[i];
 for (int i=0;i<26;i++) for (int j=i+1;j<26;j++) {
 if (!count[i] && !count[j]) continue;
 memcpy(key[n],key[0],26);
 key[n][i]=key[0][j]; key[n][j]=key[0][i]; n++;
 }
 for (int k=0;k<n;k++) { score[k]=0;
 for (int i=0;i<26;i++)
 score[k]+=count[i]*log(f[key[k][i]-'A']/100);
 }
 for (int r=0;r<top;r++) { int best=0;
 for (int k=1;k<n;k++)
 if (score[k]>score[best]) best=k;
 printf("%d: ",r+1);
 for (int i=0;c[i];i++)
 putchar(c[i]>='A' && c[i]<='Z' ?
 key[best][c[i]-'A'] : c[i]);
 score[best]=-INFINITY;
 }
}
```

OUTPUT :


Ciphertext: HHHHHH WWWWW DDDD RRR LL QQ V
Top (1-10): 3
1: EEEEEE TTTTT AAAA OOO II NN S
2: EEEEEE TTTTT AAAA OOO NN II S
3: EEEEEE TTTTT AAAA OOO II NN H
...Program finished with exit code 0
Press ENTER to exit console.